

Implementation of ICMP Based Network Management System for Heterogeneous Networks

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***Abstract** – The need for the development and use of a Network Management System (NMS) arises ipso facto that a typically huge network incorporates a huge number of interlinked network components for the transfer of data across channels making it essential to ensure the up keeping of such components for an uninterrupted communication. Such large scale deployment of components makes it tricky for an individual to pin point the fault manually within network resulting in wastage of time and money. In order to overcome these problems there is a need to develop a software that can not only automate the entire process but should also be backward compatible to support non-SNMP based devices within network avoiding the need for the deployment of new ones, hence making the entire process cost effective. The paper presents the details of an in-house developed ICMP based NMS that can perform the above tasks faster as compared to conventional approach using a multithreaded approach for both LAN and WAN.*

Keywords- Bandwidth, FCAPS, ICMP, LAN, Network Monitoring, NMS, RFC-792, WAN.

I. INTRODUCTION

Hardware and/or a software program that is used to determine and supervise the working of an individual component within network are known as NMS. An NMS can monitor various network components for malfunctioning and carry out a detailed analysis over the data so collected to determine various key aspects like responsiveness, up time, down time, average working time, packet loss, bandwidth utilization, latency etc for the device, which further helps in highlighting the faulty component(s) and the actual cause of the communication failure within network. The faults and failures analyzed after monitoring are reported to the network administrator for further action.

II. LITERATURE SURVEY

With the increasing complexity of the network due to heterogeneity, it's up keeping and maintenance is also getting complex day by day and the conventional methods to determine their operational status have really become tedious and time consuming, even the available options that have automated the process of network management are much complicated and expensive hence requiring additional skills to deploy and operate [1, 8, 9, 10]. Keeping the above factors in mind, an NMS with the ability to provide a meta-view to the users is needed that can overcome the highlighted problems and can be easily deployed and handled. For such reason an Internet Control Message Protocol (ICMP) based NMS has been developed in JAVA for heterogeneous networks that monitors & analyses the network devices both in LAN and WAN, based upon status polling implemented within multiple threads making it faster in its approach towards network monitoring and also making it suitable for old non-SNMP enabled devices due to ICMP based implementation.

A. Network Heterogeneity

A typical heterogeneous network system incorporates a large number of computers and network devices like switches, repeaters and routers working with different protocols and different operating systems with varying type, size, and topologies connected together to provide a composite service at the client end [3].

B. ICMP AND ITS WORKING

Internet Control Message Protocol (ICMP) and its functions as explained in RFC-792 [15], defines it as a protocol that is responsible for reporting the faults in datagram processing within network through the use of Internet Protocol (IP) for the purpose hence making it an indispensable part of the IP

suite [10, 15]. The various types of messages generated by this protocol are as follows:

Messages	Type
Echo Reply	0
Destination Unreachable	3
Source Quench	4
Redirect	5
Echo	8
Time Exceeded	11
Parameter Problem	12
Timestamp	13
Timestamp Reply	14
Information Request	15
Information Reply	16

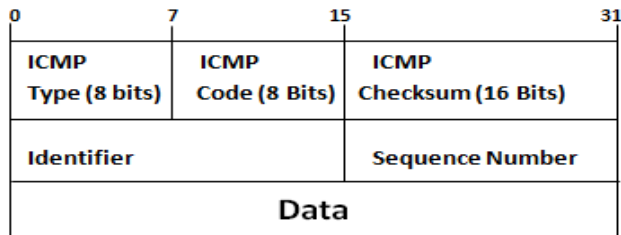


Fig.1 Basic Structure of an ICMP Header

Fig.1 illustrates the basic ICMP header format which is dependent upon the value of 'type' field [15], having a length of 8 bits, which means it can represent $2^8=256$ types of messages using the protocol with a change in the header format depending upon the value of 'type'. For the purpose of monitoring we mainly make use of ICMP type 0 and type 8, i.e. echo reply and echo respectively which helps to determine an alive or a non-alive host within a given address space. 'Ping' [18] is one of the most popular ICMP based utility provided within Windows and Linux based operating systems that are used to check the responsiveness of the network devices by forwarding an *ECHO_REQUEST* packet to the network host/gateway and receiving an *ECHO_REPLY* from them in order to determine their working status [17].

There are other suggested approaches also for the use of ICMP within network management and network discovery that can be implemented for the design and development of network management systems depending upon the network architecture and need [12, 16].

B. Important Functions of an NMS

According to ISO (International Organization for Standardization) and OSI (Open System Interconnect) framework, Network Management is divided into five broad categories (FCAPS) [3, 4, 5]:

- Fault Management
- Configuration Management
- Accounting Management
- Performance Management
- Security Management

- Fault Management
It includes recognizing the problem and reporting the same to the administrator and further tracking it till resolution.
- Configuration Management
It includes keeping the configuration backup and ensuring the health of the network.
- Accounting Management
It includes the measurement of the network utilization parameters to regulate the users for accounting.
- Performance Management
It refers to the collection of various network statistics like utilization, errors, response time in order to ensure high performance of the network.
- Security Management
It includes the various steps like authentication and authorization to protect the network from being sabotaged.

C. Properties of an ideal NMS

The ideal network monitoring/management system should have the following properties [2, 11]:

- The network monitoring should be automatic and continuous.
- The problems should be immediately reported to the administrator.
- It should be able to pin point the problem along with the location of the problem.
- It should log the changes made to the network in order to troubleshoot the problems arising later due to configuration changes.
- It should facilitate the administrator with remote authentication and authorization so that the administrator could monitor the system from anywhere.

There are various commercial and non-commercial NMS available that possess these properties completely or partially but the complications involved in their handling and maintenance make it essential to search for other cheap and easy to implement options [6, 8] .

D. NMS Design & Evaluation process

The design and evaluation process of an NMS involves the following steps [7]:

- Analyzing the characteristics and performance of the network.
- Based upon through analysis, network management objectives are proposed for the assessment of NMS.
- Simulations and application are run to evaluate the network performance and functionalities and an analysis of the developed system is done to learn about its impact on the existing network.
- Based upon all the analysis, comparisons are done to highlight the pros and cons of the developed system and

cost estimations are made for the development and deployment of the system.

- If the results are found to be satisfactory then the system is deployed with suggestions and guidance.

III. DESIGN AND IMPLEMENTATION

A. Implementation Overview

The NMS 'MAGGNET' (Monitoring Analysis & Graph Generation Network Tool) has been developed to address the problem of monitoring and management of Intranet-LAN and Intranet-WAN through an integrated user friendly user interface (UI) and to generate minimum network overhead at the same time to avoid any kind of overall network performance degradation. The software was developed in Java using NetBeans IDE 7.4 on a Windows 7 based system with 2GB RAM & Intel Core i3 processor having a clock speed of 1.8 GHz. It mainly performs status polling on regular intervals (ICMP echo request) to trigger a response from the device which is used to determine and evaluate its current and average performance rating. The software facilitates with the monitoring & manipulation of LAN & WAN simultaneously via an integrated user interface and uses an automatic Telnet based user authentication [13] process to grant access to devices within WAN hence ensuring the security of the entire network being accessed.

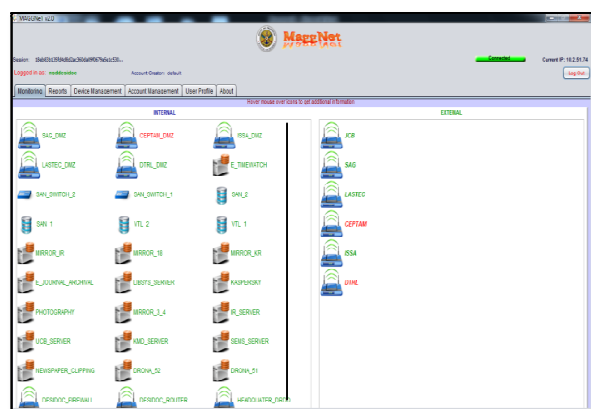


Fig.2 Screenshot of the NMS showing list of serviceable & unserviceable LAN (left) & WAN (right) devices.

Further to protect the network information from being accessed by unauthorized users, the software provides a three level user authorization mechanism which implements salted SHA-512 hash function making it secure against most of the cryptanalytic attacks [14] hence adding to the security of the entire network as well as the system.

B. Device Discovery

Device Discovery and topological reconfiguration are the most important part of configuration management which includes timely and automatic updating of the topological view of the existing network upon any kind of alteration in the physical

network architecture, especially for the devices like routers that serves as a link for various other devices. In our approach towards automatic device discovery the above goal is achieved by transmitting the ICMP packets in broadcast mode with the address FF:FF:FF:FF:FF:FF (hexadecimal), so that the packet is simultaneously received by all the nodes of the network hence reducing the load on the network management system and increasing the speed of discovery process. Once the ICMP echo reply indicating a live node is received at the transmission end, the device is added to the topology and the corresponding information is saved to a centralized database. To avoid any inconsistency in topological view upon manual addition or deletion of any node, a module compares the number of nodes in the current topological view and the database to trigger an automatic device discovery/ refresh process that recreates the new topological view over the screen if a mismatch is found in the number of elements in both the lists according to the following pseudo code:

```
int countNodes1:=count(ElementsInDatabase);
int countNodes2:= 0;
void refresh( )
{ if (countNodes2<countNodes1) then
  { // discover devices for first launch
    // create topological view
  }
  elseif (countNodes2>countNodes1) then
  { //device deletion has occurred
    //recreate topological view and update database
  }
  countNodes2:=countNodes1.
}
```

In the above pseudo code countNodes1 and countNodes2 are the global variables and the function refresh() is called whenever a node is deleted, after subtracting 1 from the current value of countNodes1.

C. Monitoring Process (Status Polling phase)

The topological reconfiguration is followed by the monitoring process which includes continuous status polling on predefined interval. In order to achieve higher throughput from the software, a multithreaded approach is being adopted towards status polling of the network devices, where a list of three IP addresses is maintained at a time to initiate polling, after testing the status of the devices in the list, next three elements are fed to the list using a loop and all the threads running the status polling module are managed by a scheduler. The number of threads for monitoring can be increased or decreased depending upon the requirement and as supported by the network and other resources to keep the entire system to an optimum state. The general steps involved in the monitoring process are as follows:

1. Insert IP address into polling list.
2. Send ICMP *echo_request* (type 8) packets to the network device according to the list.

3. Receive ICMP *echo_reply* (type 0) packet from the device.
4. Extract the operational status information from the packets.
5. Get the success threshold value for status determination of the device.
6. Compare the operational status information with the threshold value and highlight the devices using color code on the screen.

As the suggested NMS works for both LAN and WAN, the status polling process for the network devices under WAN requires to be accessed through a telnet client based user authentication process which is being implemented within software using *org.apache.commons.net.telnet.TelnetClient* java package. Under this process a secured connection is established to the WAN network by connecting to the server on port 23 using password, if the connection succeeds then monitoring of devices is started for the WAN otherwise a connection failure message is prompted. Fig.3 shows the general approach towards WAN monitoring.

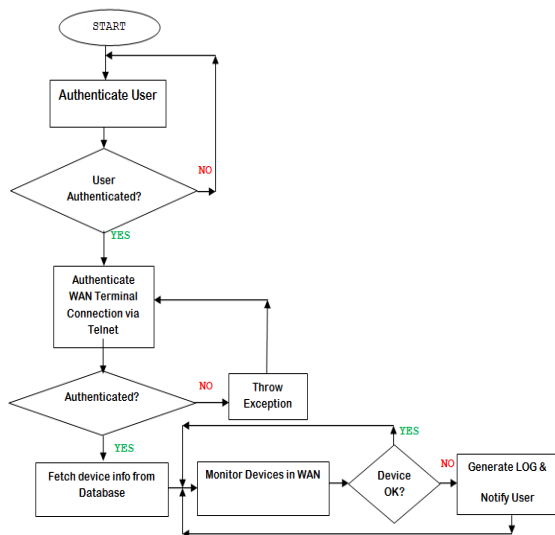


Fig.3 Flowchart of the Intranet WAN monitoring process

The code to established secured terminal connection for monitoring WAN devices is implemented in java in the following way:

```

import org.apache.commons.net.telnet.TelnetClient;
public class AutomatedTelnetClientServer {
    private TelnetClient telnet = new TelnetClient();
    private InputStream in;
    private PrintStream out; private String prompt = ">";
    public AutomatedTelnetClientServer(String serverIP, String password) {
        try { telnet.connect(serverIP, 23);
            in = telnet.getInputStream();
            out = new PrintStream(telnet.getOutputStream());
            readUntil("Password: ");

```

```

            readUntil(prompt);
        } catch (Exception e) {e.printStackTrace();}
    }
    public String sendCommand(String command) {
        try {write(command);
            return readUntil(prompt);
        } catch (Exception e) {e.printStackTrace();}
        return null;
    }
    public void disconnect() {
        try {telnet.disconnect();
        } catch (Exception e) {e.printStackTrace();}
    }
    AutomatedTelnetClientServer telnetdev = new
    AutomatedTelnetClientServer(<IP address>, <password>);
    // Send echo_request(); and analyse response.
    telnetdev.disconnect();
    In the above code, the method
    AutomatedTelnetClientServer(<IP>, <password> ) accepts
    the IP and password as the parameter to establish connection
    to port 23 of the remote server and shows the prompt in case
    of success and a failure message in case of connectivity
    failure.

```

D. Performance Analysis of the Network Devices

The performance and fault analysis are the part of monitoring process that takes the input from the output of the status polling phase and initially calculates the average latency and the packet loss for the device. Further it acquires the threshold value for latency and the packet loss and performs a comparative analysis to determine the health and performance of the device.

We assume that 'latency' is the calculated average latency and 'loss' is the calculated packet loss for some device under monitoring, also we assume that the threshold value of average latency for a LAN as well as WAN is 15ms (considering the network of DESIDOC, DRDO), then the performance of the device is determined according to the following pseudo code.

```

If (loss=100% OR latency>15) then
    {Highlight the device with red color and raise alert with
    relevant information.}
Else If (latency<15 && (loss < 100 && loss>=75) then
    {device is showing poor performance and must be highlighted
    with orange color.}

Else if (latency<15 && (loss < 75 && loss>=50) then
    {Minor network fault. Notify admin.
    //other action if desired.}
Else if (latency<15 && (loss < 50 && loss>=25) then
    {Show Device yellow.
    //other action if desired.}
Else if (latency<15 && (loss < 25) then
    {Device in excellent working condition.
    Show device green.}
If (loss> threshold packet loss value) then

```

```
//Trigger alert
//Take necessary actions
}
```

The packet loss is calculated by dividing the number of ICMP echo_reply packets received by total number of ICMP echo_request packets sent to the device during status polling and multiplying the entire value with 100.

The Java code (method) implemented within software for calculating packet loss is as follows:

```
public int GetPktLoss(String ip)
{String [] pkt=null; String PingOutput;
/* echo_request is a method to send ICMP packets to the
network device using ip and the number shows the number of
packets sent. */
try {Output=this.echo_request(ip, 4);
pkt=Output.split("s"); pkt=pkt[1].split(",");
int sent= Integer.parseInt(pkt[0]);
pkt=Output.split("r");pkt=pkt[1].split("//");
int received= Integer.parseInt(pkt[0]);
} catch (Exception e){ System.out.println(e);}
Return ((received/sent)*100);
}
```

echo_request() is the function associated with the status polling phase that sends the custom sized and number of ICMP packets to the designated IP address and returns the response from the device in the format (s <no. of packets sent >, r <no. of packets received>) eg. (s4, r3) means 4 packets were sent and 3 were received. Further the function GetPktLoss() calculates the drop in the received packets and returns the loss percentage.

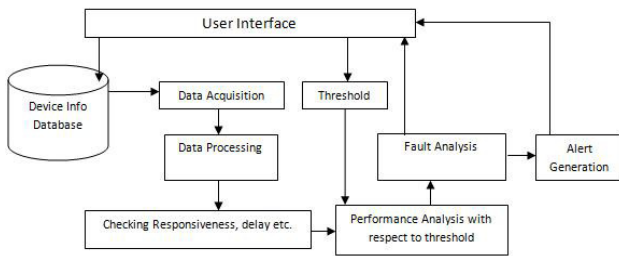


Fig.4. Schematic diagram of the overall system

IV. RESULTS

The implementation shows that when the status polling of the device takes place sequentially for a large number of devices using a single thread, the time consumption is nearly proportional to the number of devices, whereas by implementing a multithreaded approach the time required for status polling is reduced considerably as compared to single threaded approach implemented by various available NMS.

It shows almost constant behavior for a small increase in the number of devices when the actual range is very large as shown in Fig. 5.

Fig.6 shows the test results for the network usage by the developed software for 45 most significant network devices within campus of Metcalf House, DRDO, Delhi (including switches, servers, routers, etc.) which displayed an average value of 26.3Kbps on a typical Gigabit Ethernet LAN and WAN when polling interval was set to 3 minutes.

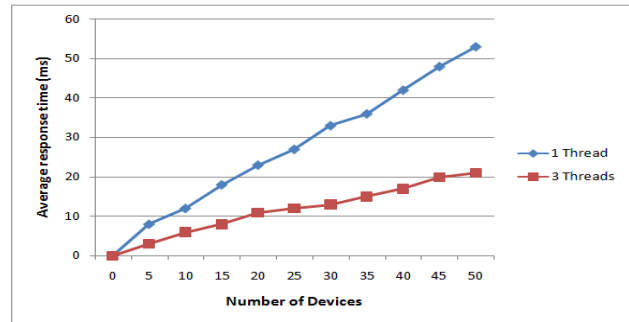


Fig.5. Comparative analysis of the single threaded approach Vs multi threaded approach

Further use of broadcast mode for device discovery increased the speed of the device discovery process as compared to discovery of devices one by one using ICMP. Also it worked well for both SNMP and non-SNMP based devices irrespective of the network heterogeneity.

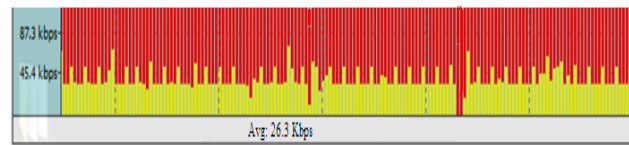


Fig.6. Screenshot of network usage by the NMS

The network usage includes the ICMP echo requests made by the software to get the status of the network resources as well as the total data transfer including the storage at the centralized database.

Using salted SHA-512 and AES-128 for data storage and message transfer makes the system secured against most of the cryptanalytic attacks [14].

Optimum performance by the software was shown when tested upon systems with following configuration:

TABLE I. SYSTEM CONFIGURATIONS USED TO TEST THE SOFTWARE

OS	RAM	Processor	Clock Speed
Windows Vista (32 bit)	2 GB	Pentium Dual Core	1.73 GHz
Windows 7 (32Bit)	1 GB	Intel Core i3	1.8 GHz

Windows 7 (64 Bit)	4 GB	Intel Core i3	1.8 GHz
Windows 8.1 (64 bit)	4 GB	Intel Core i5	2.8 GHz

The memory usage analysis of the developed software suggests that the minimum memory required by the software on a 32 bit Windows Vista (2GB RAM, Pentium Dual Core@ 1.73GHz) based standalone system while in execution is 21 MB.

V. CONCLUSION AND FUTURE SCOPE

In this paper we have presented the design and implementation strategy for a pure ICMP based NMS that satisfies most of the requirements of a typical NMS and provides a unified interface for monitoring and management of network resources. Further on the basis of information acquired by running multiple simulations over network suggests that due to multi threaded approach for status polling makes the software perform faster as compared to other ICMP based NMS and also the broadcast mode device discovery consumes very less network resources hence avoiding the overall network performance degradation. The suggested approach displayed optimum performance for both LAN & WAN. The future enhancement includes the implementing SNMP within software and enhanced interface for remote device management.

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